
Launch of SpaceX Falcon 9 with Dragon capsule for re-supply missions

Blue Origin's New Shepard suborbital system

just bounce off the atmosphere or break up. Furthermore, as the capsule carrying humans enters the atmosphere, it is in free fall, accelerating tremendously. The sheer velocity of the entering craft is so fast that the air molecules can't move away in time, and this causes such high pressures, that the air is turned into plasma (the fourth state of matter). Space craft have heat shields to keep the colossal amounts of heat from getting into the craft. A part of the heat-shield failed for the Columbia, and it's sad that they were literally vaporised!

However, that disaster made us more careful, and all the buzz these days is about the rockets that will carry us into space. There just aren't enough things going up into space, because there aren't enough affordable rockets, and there aren't enough affordable rockets because there aren't enough things going up into space.

## Private vs. Government

For decades, space exploration and research was in the hands of governments the world over due to the prohibitive costs for going into any mission – in the range of billions of dollars.

Businesses are not run by incurring huge costs, coupled with tremendous risks, and no real business model of returns. That's not something any shareholder of a private company is going to give a go ahead to. Countries, on the other hand, are run by politicians, who have very different agendas, and are willing to loosen the purse strings for non-monetary gain, such as national pride, and winning the Cold War, for example. This is why countries push the frontiers, and companies make it cheap and profitable once all the risks are known.

Today, Low Earth Orbit space travel has known risks, and that's why private companies are needed to take the baton over from countries, to do what they do best – compete, find ways to bring down costs, and invent new business models.

While countries push the frontiers even further, it's up to the private companies to make the known frontiers more efficient. Why wouldn't NASA or ISRO use a private company to launch their satellites or payloads when a private company can do it at half their cost? Since the biggest clients for private companies will be the government programs themselves, we're at a point where they will all work in a symbiotic relationship.

## The new entrants

Every private space launch company out there is aiming to bring launch costs down for rockets. The most well-known name is obviously Elon Musk's SpaceX, which has set the ultimate goal for themselves of colonising Mars.

SpaceX has set all kinds of records: their Dragon was the first private craft to launch, orbit, dock with the ISS, and be recovered; Falcon 1 was the first private rocket to launch into orbit; Grasshopper was the first technology demonstrator to take off vertically and land vertically (VTVL) from a short height, and the mighty Falcon 9, the first private rocket to attempt a vertical landing after launching into space, failing soft landing thrice before finally succeeding this month.

The other big name suddenly is Jeff Bezos's Blue Origin. Taking the whole